

## Project overview

### Title: InternSmart

## 1. Project description

InternSmart is a comprehensive web-based platform designed to manage the complete academic internship lifecycle from account registration to the final oral defence simulation. The system brings together students, academic supervisors, professional supervisors, and administrators in a single unified digital workspace.

The platform integrates two AI models: Google Gemini 1.5 Flash for report quality analysis and theme extraction, and GPTZero for AI content detection alongside a live peer-to-peer video system powered by WebRTC, a real-time notification engine using Socket.io, and a structured PostgreSQL database. Together these technologies automate key aspects of internship supervision while keeping human judgment at the center of every important decision.

The system addresses a gap that no existing platform currently fills: a unified solution covering report submission, AI-powered quality review, dual supervisor workflow, academic integrity monitoring, and live defence simulation all within one coherent platform specifically designed for the academic internship period.

## 2. Problem statement

The management of the academic internship period currently suffers from six identified problems:

- ❖ Absence of a centralized platform: report submissions happen through email or physical handover with no traceability, no version history, and no structured record of exchanges between students and supervisors.
- ❖ No automated report quality control: supervisors spend significant time correcting basic writing errors instead of focusing on deep academic and professional evaluation.
- ❖ Rise of AI-generated content with no detection mechanism: the widespread use of AI tools like ChatGPT makes it difficult to assess the authenticity of student work, and no integrated detection mechanism exists within current internship workflows.
- ❖ No structured defence preparation support: students receive no formal tool to practice their oral defence in a simulated environment before the actual defence date, leading to poor verbal performance despite good written work.

### 3. Overview of existing system

#### 1 Overview of Existing Tools

Several platforms currently exist that address aspects of academic management or internship supervision. However none of them was designed specifically for the academic internship lifecycle, and none provides a unified solution covering all the challenges that students, supervisors, and administrators face during this period. The table below presents the most widely used existing tools alongside their capabilities and critical limitations.

| System                   | What It Does                                                                                | Critical Limitations                                                                                                                                |
|--------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Google Classroom         | Document submission, assignment management, basic feedback between teachers and students    | Not designed for internship context. No AI report analysis. No AI detection. No defence preparation. No dual supervisor structure. No live session. |
| Moodle                   | Learning management, document submission, forums, quizzes, grade tracking                   | Generic LMS not adapted to internship supervision. No AI integration. No version-controlled report submission. No defence simulation module.        |
| Turnitin                 | Plagiarism detection in academic documents. Recently added basic AI detection.              | Standalone tool only. Not integrated into any supervision workflow. No report quality feedback. No feedback mechanism. No defence preparation.      |
| GPTZero / Originality.ai | Specialized AI content detection tools that return probability scores for AI-generated text | Standalone detection tools with no connection to any academic workflow. No report submission. No supervisor interface. No defence module.           |
| Handshake / Symplicity   | Internship placement platforms matching students with companies for recruitment             | Focus on job placement only. No report submission. No supervisor review workflow. No AI quality analysis. No defence preparation.                   |
| Academic Interlocutor    | AI chatbot that simulates oral exam questions to help                                       | Standalone chatbot only. No webcam. No real supervisor. No report context. Not connected to                                                         |

|                        |                                             |                                                                                                                                |
|------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                        | students practice articulating knowledge    | any internship management workflow.                                                                                            |
| Microsoft Teams / Zoom | Video conferencing for meetings and classes | No internship-specific features. No report submission. No AI analysis. No structured defence workflow. No assignment tracking. |

## 2 The Gap

As demonstrated in the table above, existing tools address at most one or two aspects of the internship management challenge. Some handle document submission but not AI analysis. Some detect AI content but are disconnected from any supervision workflow. Some offer video calling but have no internship-specific context or structure. None of them combines all of the following in a single platform:

- Centralized report submission with version control.
- Automated AI quality review with highlighted errors and suggestions.
- AI content detection integrated directly into the submission pipeline.
- A two-stage submission flow where the student works with AI before involving supervisors.
- A dual supervisor structure with role-based filtered views enforced at the server level.
- Supervisor-controlled defense questions based on the approved report.
- A live peer-to-peer defense simulation session between the student and their supervisor.
- Session scheduling to handle supervisor availability.
- An end-of-period AI usage report for institutional oversight.

## 4. Project objectives

The general objective of this application is to design and implement an intelligent web platform that automate and modernizes the academic internship supervision process by integrating artificial intelligence and live video communication technologies.

The primary objective of this project is to design and develop a web-based platform that integrates Artificial Intelligence to achieve the following specific goals:

- Provide a centralized and secure platform for students to submit internship reports and for supervisors to review and respond with structured feedback.

- ✚ Implement a two-stage submission flow where the student works with AI to improve their report first before sending it to supervisors for review.
- ✚ Integrate AI-powered report quality analysis using Google Gemini 1.5 Flash to identify errors, suggest corrections, and extract the project theme automatically.
- ✚ Embed AI content detection using GPTZero to calculate and report the proportion of AI-generated content in each submission, visible to supervisors and the admin.
- ✚ Support a dual supervisor structure with role-based filtered views academic supervisors see full AI grammar analysis while professional supervisors see only the AI detection percentage.
- ✚ Enable the academic supervisor to write personalized defence questions based on the approved report and conduct a one-time live defence simulation session with the student via WebRTC peer-to-peer video.
- ✚ Implement an automated notification system that alerts all users about deadlines, feedback, and session readiness via real-time in-app alerts and email.
- ✚ Allow administrators to manage user accounts, activate registrations, set the internship timeline, and generate an end-of-period AI usage report across all students.

## 4. In Scope

- Self-registration for students and supervisors with admin activation workflow.
- Student-supervisor linkage established by the student during registration based on externally assigned supervisors.
- Report submission with version control and AI analysis pipeline.
- Dual supervisor review with role-based filtered views enforced server-side.
- Defence preparation module with supervisor-written questions and live WebRTC session.
- Automated notifications via Socket.io and Nodemailer with daily deadline reminders via node-cron.
- Admin AI usage report with filtering and export capabilities.

## 5. User and their roles

| User role           | Responsibilities                                                                               | What they can see                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Student             | Register, submit reports, work with AI, send to supervisors, attend live defence session       | Own dashboard, own reports, AI feedback, both supervisor feedback, defence questions and session |
| Academic Supervisor | Review reports, send academic feedback, approve reports, write defence questions, conduct live | Assigned students only, full AI grammar analysis, AI detection %, defence session interface      |

|                         |                                                                           |                                                                                        |
|-------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
|                         | session, give defence feedback                                            |                                                                                        |
| Professional Supervisor | Review reports, send professional feedback, approve reports               | Assigned students only, AI detection % only (grammar analysis hidden), report document |
| Admin                   | Activate accounts, set timeline, view AI usage report across all students | All users, all stats, AI detection report for all students, system activity            |

## 6. Key features

### 6.1 Self-Registration with Admin Activation

Students and supervisors register themselves on the platform. All accounts start as pending until the administrator reviews and activates them. This ensures full institutional control over who can access the system.

### 6.2 Two-Stage Report Submission Flow

The student submits their report and works privately with the AI to improve it through as many correction cycles as needed. Only when satisfied does the student click a separate button to send the report to both supervisors. This means supervisors always receive pre-corrected, higher quality reports.

### 6.3 AI-Powered Report Quality Analysis

Google Gemini 1.5 Flash analyzes the submitted report text and returns a structured JSON response highlighting grammatical errors, structural weaknesses, repetitions, and missing sections each with the exact location, explanation, and a concrete suggestion. It also extracts and stores the project theme for later use in the defense module. An overall quality score out of 10 is calculated.

### 6.4 AI Content Detection

GPTZero analyzes the report text and returns an AI detection percentage based on perplexity. The result is displayed to both supervisors and stored in the database for the admin's end-of-period report. It is presented as a supplementary indicator, not a definitive verdict.

### 6.5 Dual Supervisor Review with Role-Based Filtered Views

Both supervisors independently review the same report from their own dashboards. The academic supervisor sees the full AI grammar analysis. The professional supervisor sees only

the AI detection percentage. This filtering is enforced server-side through JWT role verification. it cannot be bypassed from the frontend.

### 6.6 Live Defence Simulation via WebRTC

After both supervisors approve the report, the academic supervisor writes personalized defence questions. The student prepares privately offline. When ready the student initiates a one-time live session. Both connect via peer-to-peer WebRTC video directly in the browser. no external software needed. The supervisor conducts the session live, listens to the student's answers, and writes detailed feedback afterward.

### 6.7 Automated Notification Engine

Socket.io delivers real-time in-app notifications to online users instantly. Nodemailer sends email notifications for all critical events. node-cron runs daily deadline checks and automatically sends reminders to students and supervisors as deadlines approach.

### 6.8 Admin AI Usage Report

At the end of the internship period the administrator generates a complete report showing every student's AI detection percentage, class, and academic supervisor. Results are filterable by class and risk level (low, medium, high) and exportable as CSV.

## 7. Technology stack

| layer             | Technology                     | Purpose                                                    |
|-------------------|--------------------------------|------------------------------------------------------------|
| Frontend          | React.js                       | User interface, role-based rendering, real-time updates    |
| Backend           | Node.js + Express.js           | API server, business logic, external API orchestration     |
| Database          | PostgreSQL                     | Relational data storage                                    |
| Authentication    | bcrypt + JWT                   | Password hashing and token-based access control            |
| File Handling     | multer + pdf-parse + mammoth   | Upload, storage, and text extraction from PDF              |
| AI Quality Review | Google Gemini 1.5 Flash (free) | Error detection, writing quality scoring, theme extraction |
| AI Detection      | GPTZero API (free tier)        | AI content detection and percentage scoring                |
| Live Video        | WebRTC                         | Peer-to-peer live video session for defence simulation     |
| Real-Time         | Socket.io                      | In-app notifications and WebRTC signaling                  |
| Email             | Nodemailer + SMTP              | Email notifications for all critical events                |
| Scheduling        | node-cron                      | Automated daily deadline reminder checks                   |

## 8. innovations

- Two-Stage Submission Flow: Student corrects with AI privately before supervisors see any version, a design that improves report quality before human review begins.
- Dual Supervisor Structure with Server-Side Filtered Intelligence: Two supervisors with different professional angles evaluate the same report through role-filtered views enforced at the backend level.
- AI Detection Embedded in the Supervision Workflow: Not a standalone tool but an automatic step in every submission, with results visible to all evaluators and aggregated for institutional oversight.
- Supervisor-Controlled Defence Questions: The academic supervisor retains full intellectual control over what is asked. The AI assists only with report analysis — it never generates questions.
- Live Peer-to-Peer Defence Simulation: One real session between student and supervisor built inside the platform using WebRTC. the student has one chance, mirroring real defence pressure.
- End-to-End Internship Lifecycle: The only platform in the world covering registration, AI-assisted report submission, dual supervisor review, live defence simulation, and institutional AI integrity reporting in one unified system.

## 9. expected outcomes

- A fully functional and deployable web application covering the complete academic internship lifecycle.
- Significant reduction in supervisor workload through automated AI pre-screening of reports.
- Improved report quality as students receive detailed AI feedback before human review.
- Transparent academic integrity monitoring with AI detection results aggregated at the institutional level.
- Better-prepared students for their oral defence through a realistic live simulation with their actual academic supervisor.
- A centralized, traceable, and auditable record of all internship activities, submissions, feedback, and defence evaluations.

## 10. CONTEXT

The management of internship period represents one of the most decisive phases in the training of a software engineering student. It is during this period that the students leave the theoretical environment of the classroom and confront real professional challenges within a company. At the end of this experience, the student must produce a written report summarizing their work and defend it orally before a jury panel composed of academic evaluators.

This transition from theory to practice is essential and valuable. However, the institution responsible for supervising this process are forced with increasing challenges. The number of students enrolled in internship programs is growing every year. In a single institution, hundreds of students may be completing internships simultaneously, each requiring individual supervision from both an academic and the professional supervisor. Managing this volume of supervision with tools currently has become increasingly inefficient.

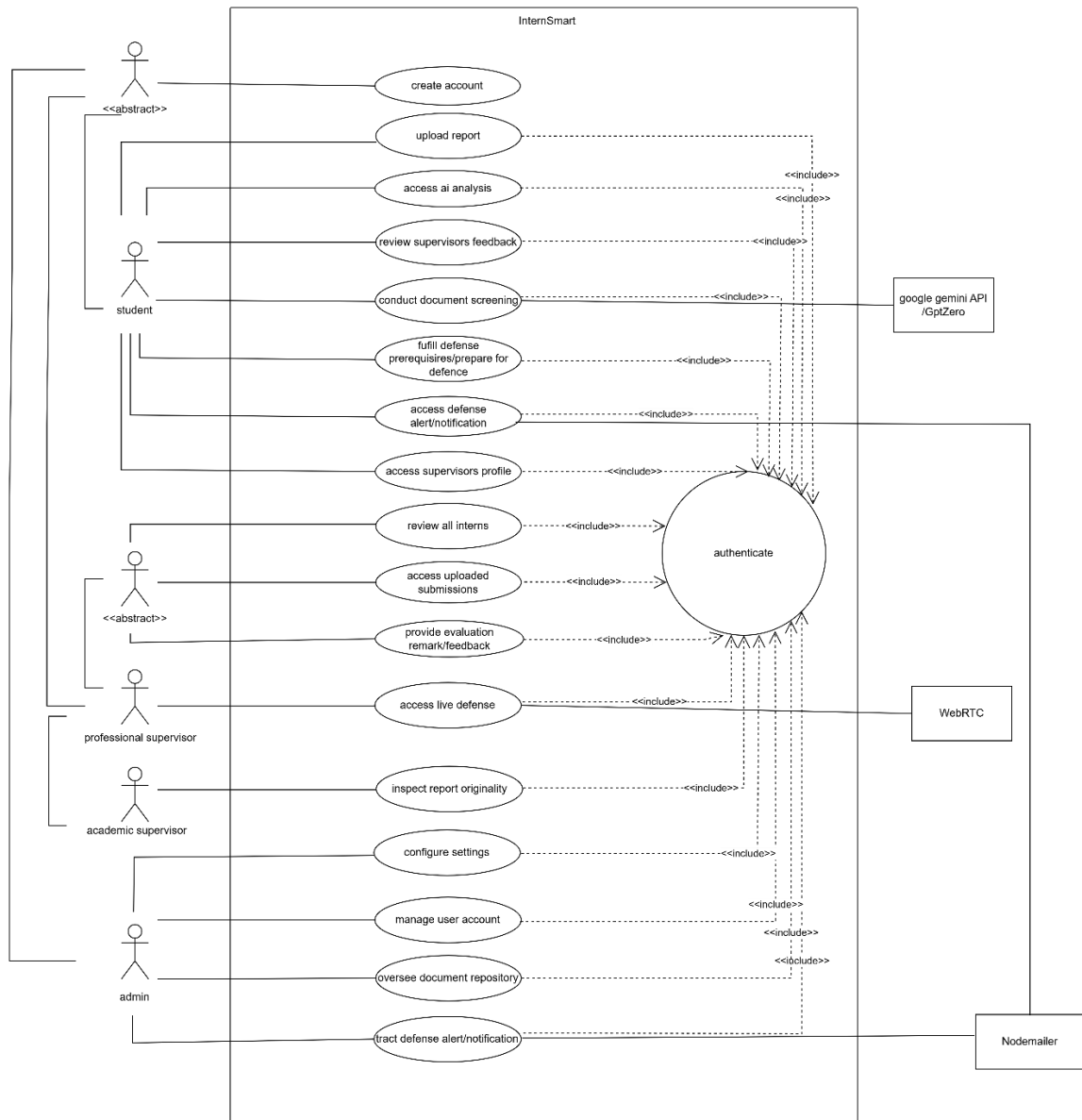
Furthermore, the emergence of powerful AI writing tools such as CatGPT, Gemini and others has fundamentally changed the academic landscape. Students now have access to tools capable of generating complete, well written academic texts in seconds. This development raises serious questions about the authenticity of student work and ability of supervisors and institutions to verify whether a submitted report genuinely reflects the students own understanding and effort.

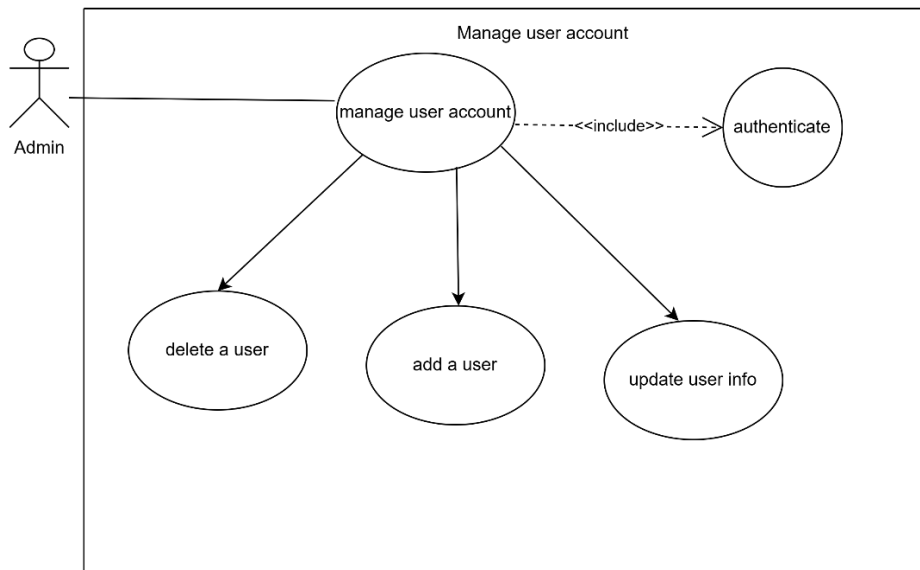
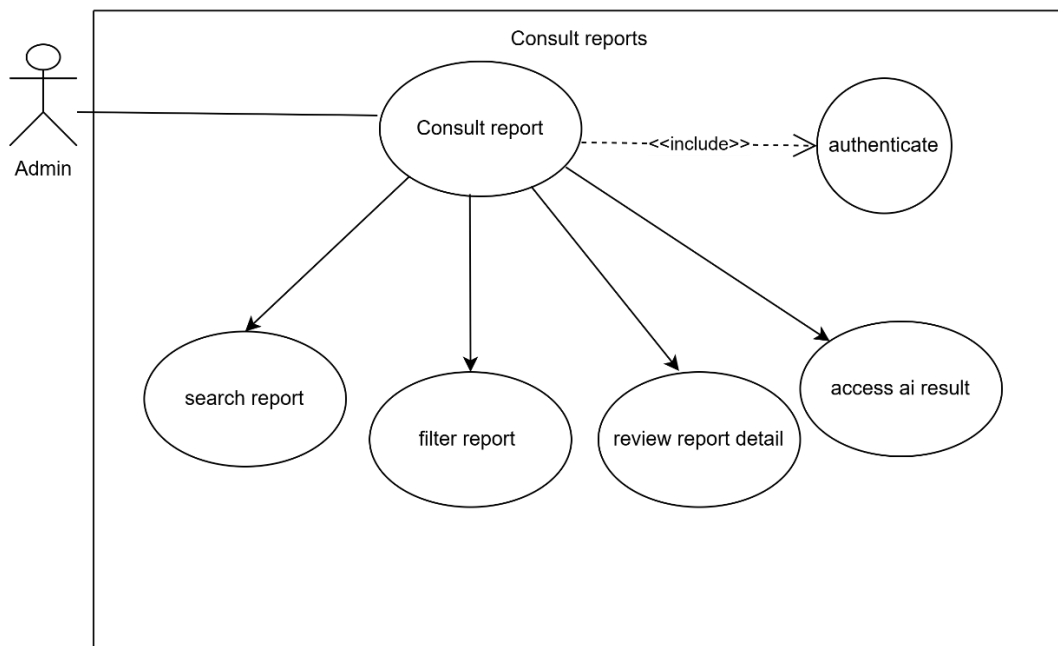
At the same time, the preparation of students for the oral defence remains largely informal and unstructured. Most students receive no guided practice before standing Infront of the jury. The quality of a student's oral performance on defence day often bears little relationship to the quality of their written work simply because they had no opportunity to rehearse. It is within this context: Growing supervision complexity, rising AI assisted academic dishonesty, and inadequate defence preparation that the present project was conceived.



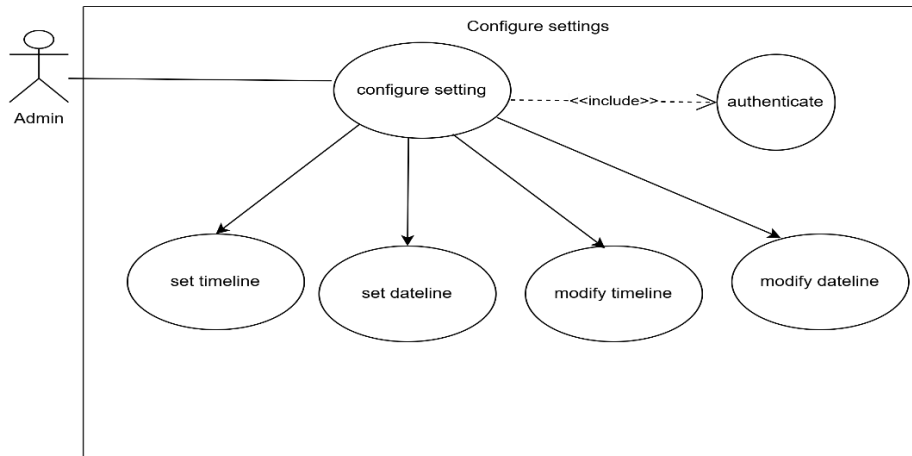
# 11. Behavioural diagrams

## 1. Use case diagram

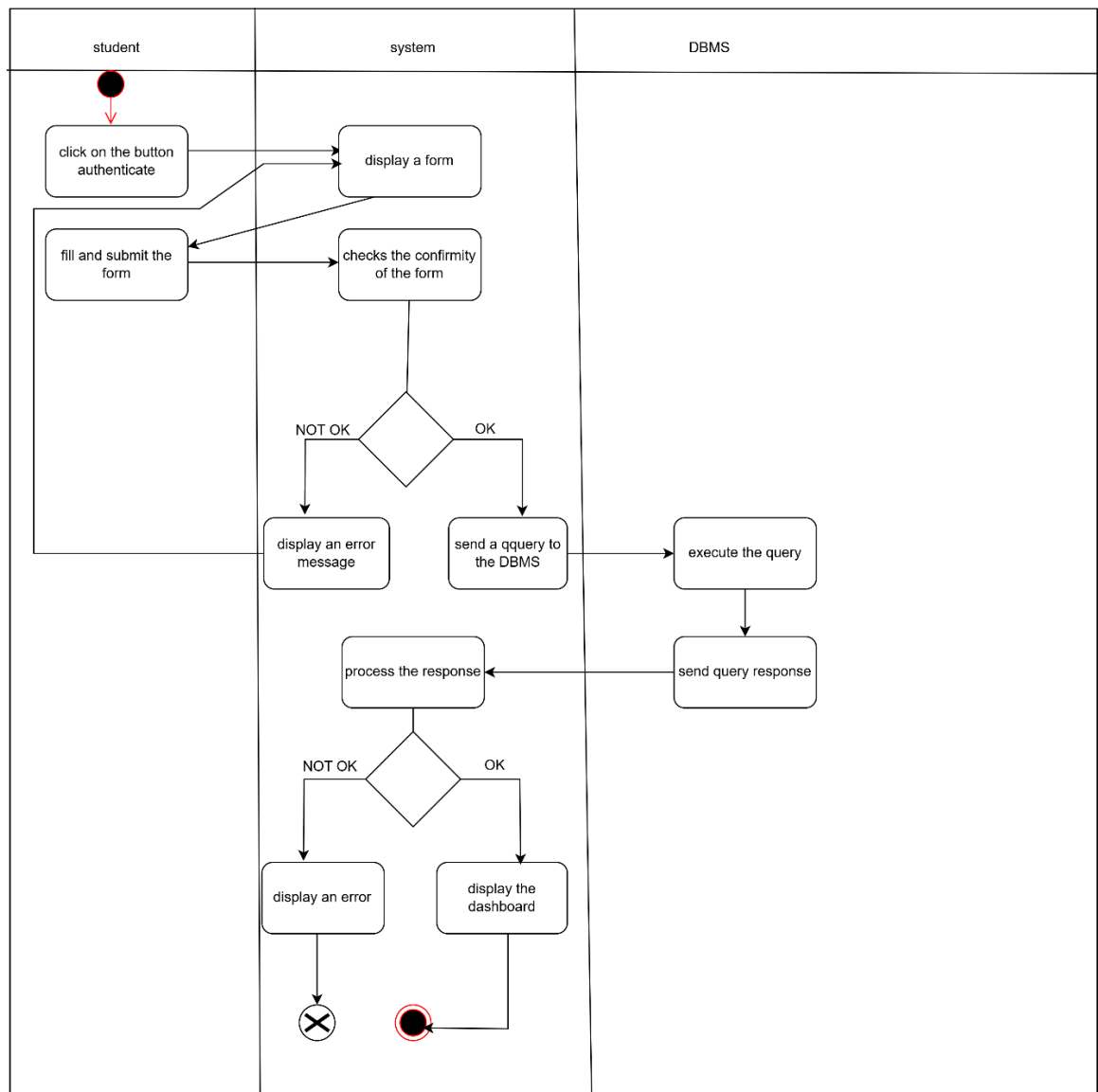


**Specific use cases****1.1 Manage user account****2.2. consult all reports**

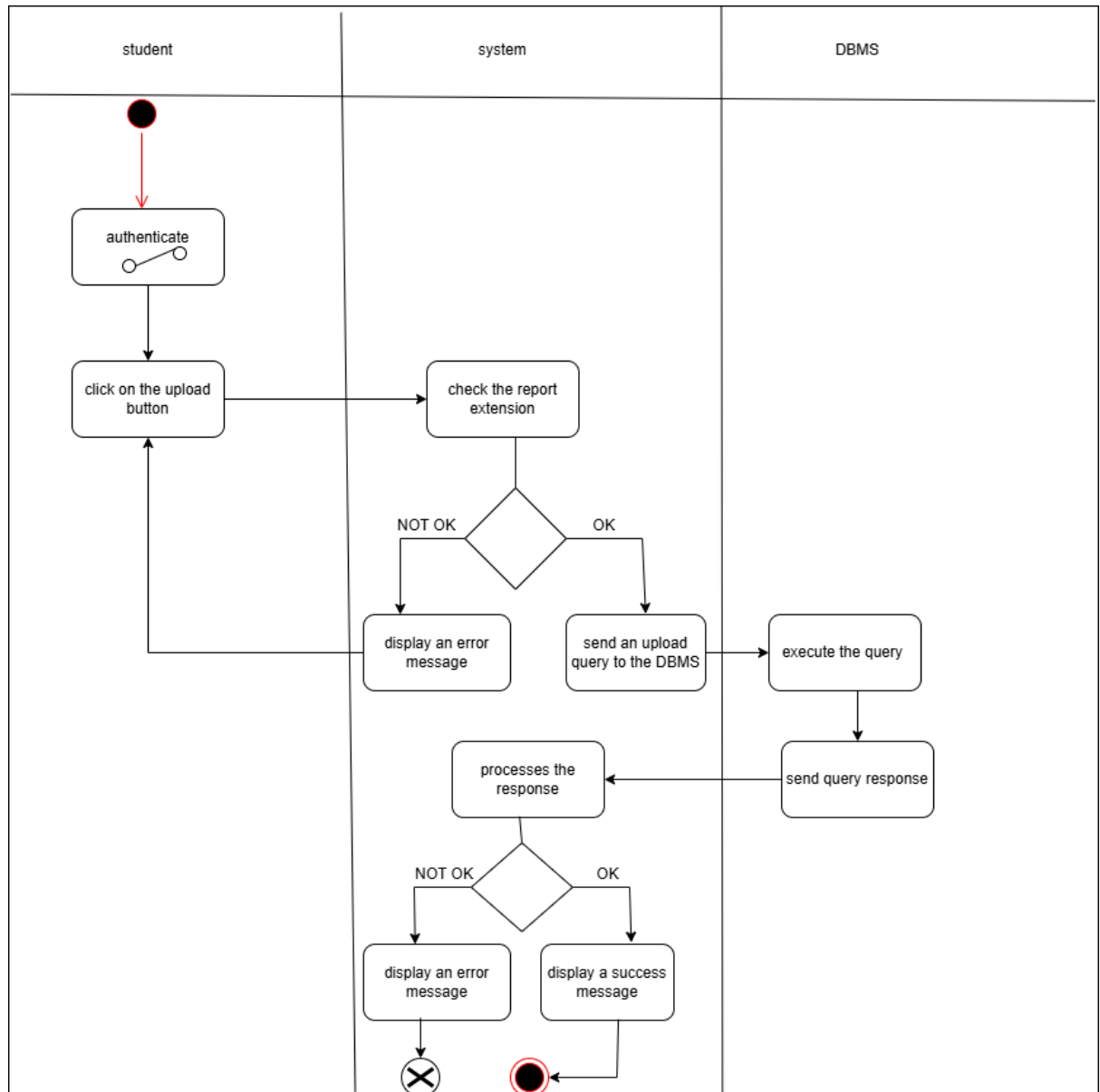
## 1.2 Configure settings



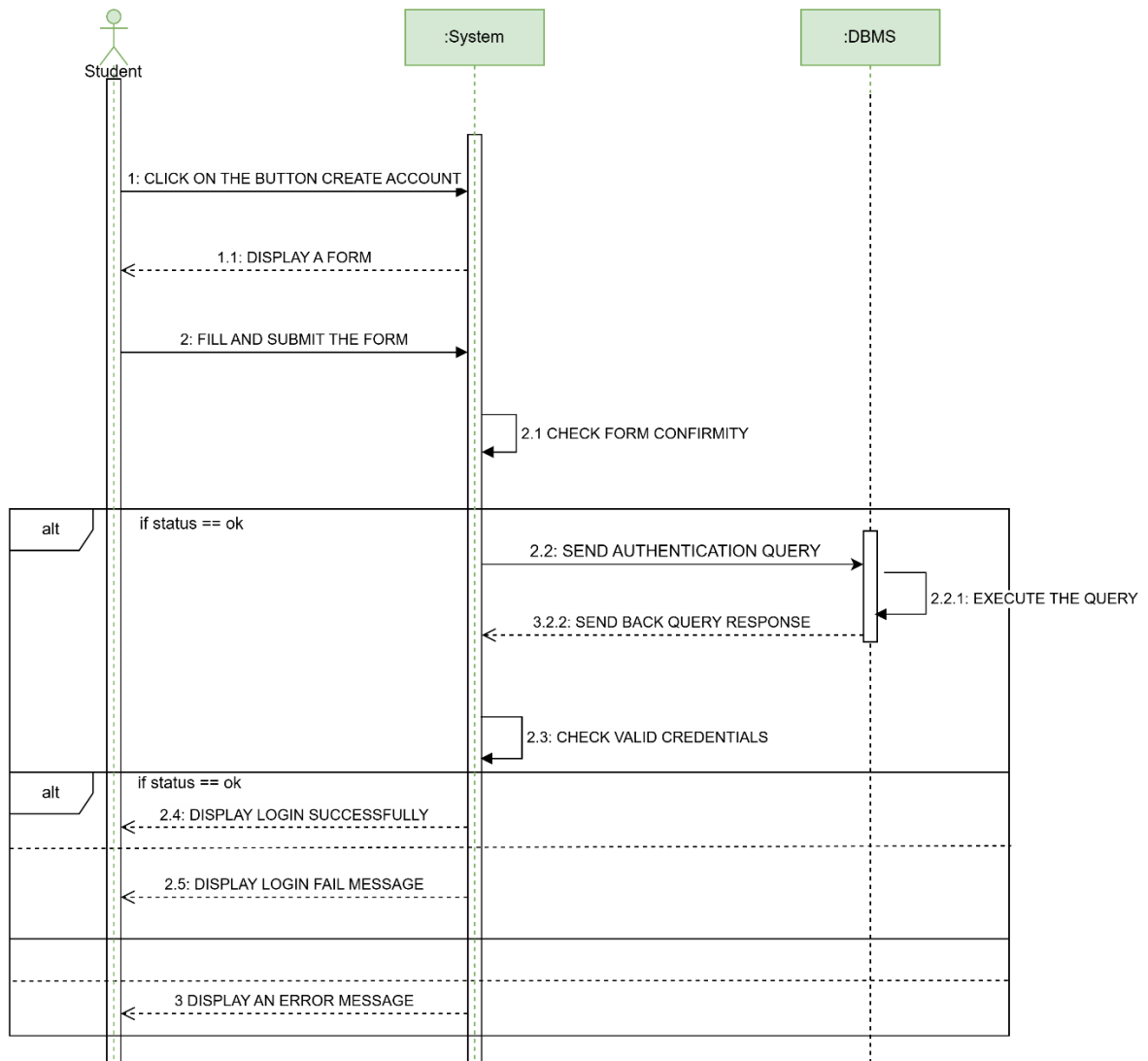
## 2. Activity diagram to authenticate



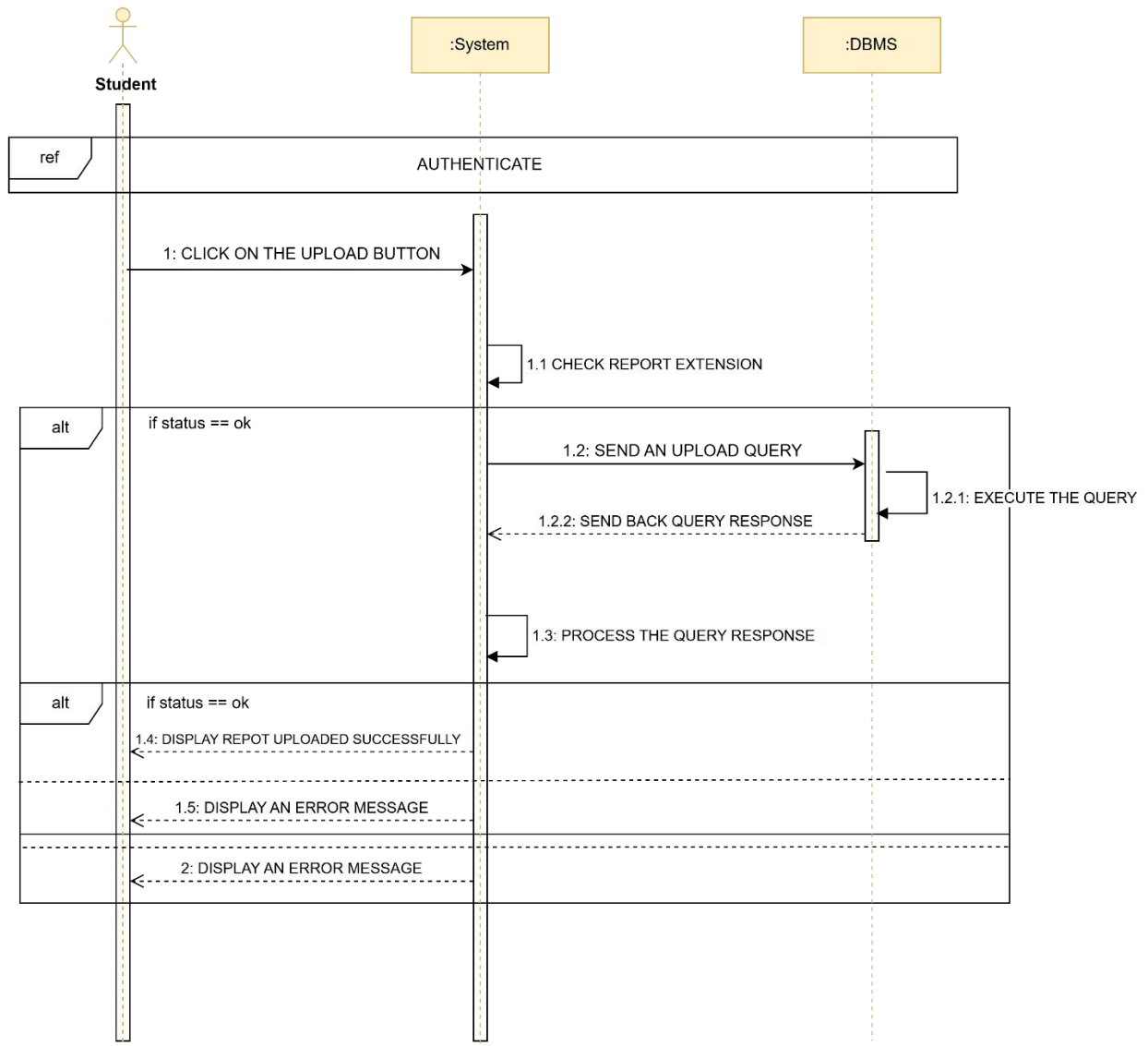
## 2.2. Activity diagram to upload report



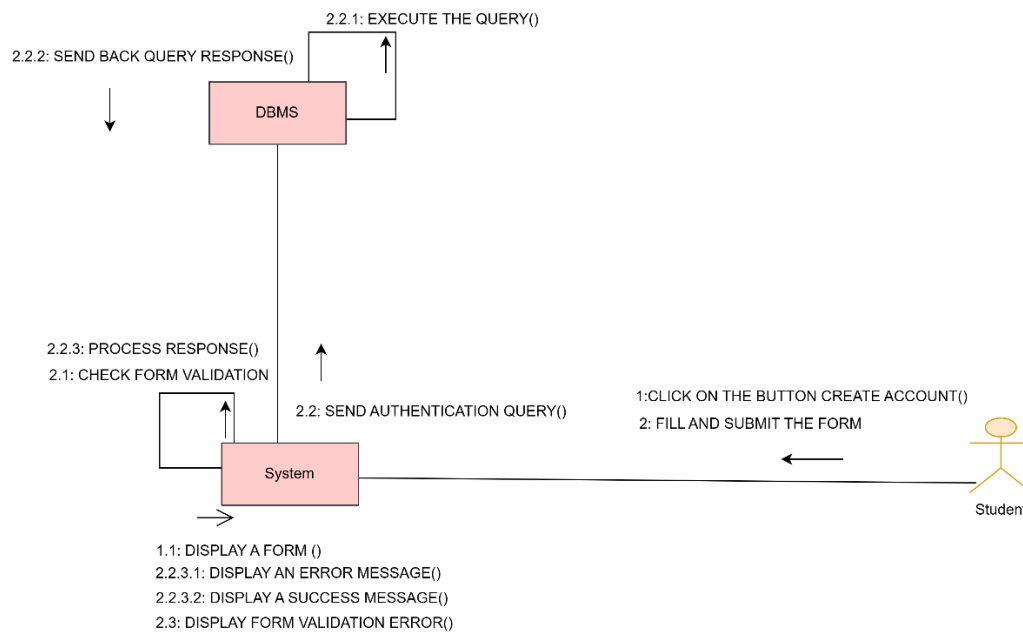
### 3. Sequence diagram to authenticate



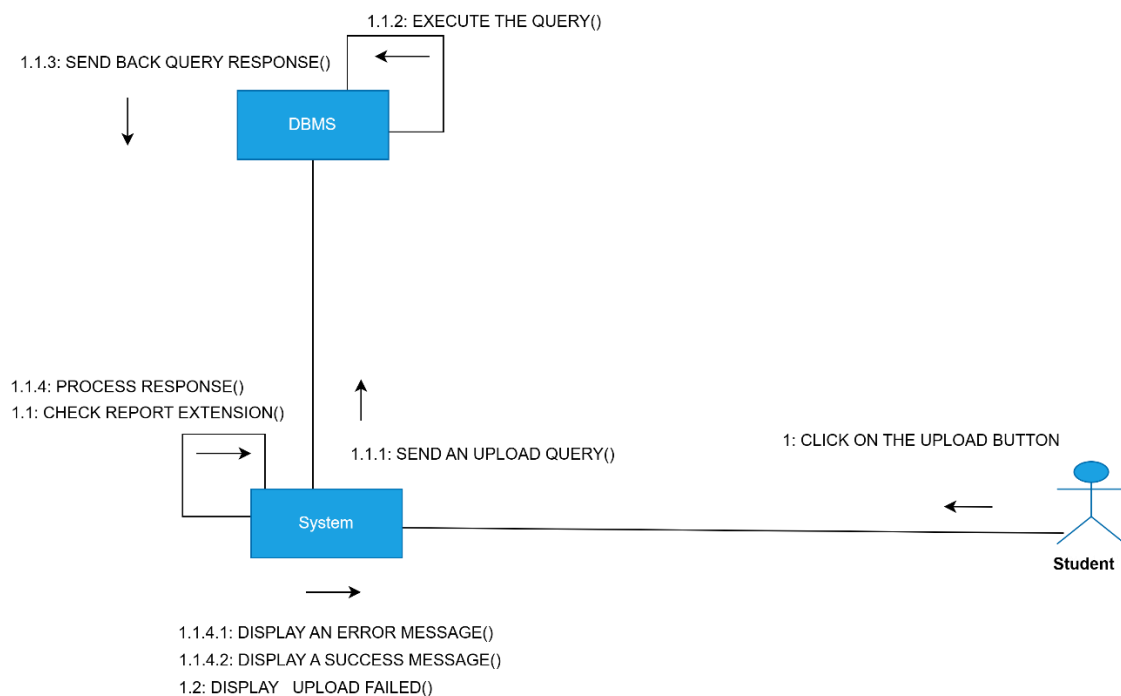
## 3.1 sequence diagram to upload report



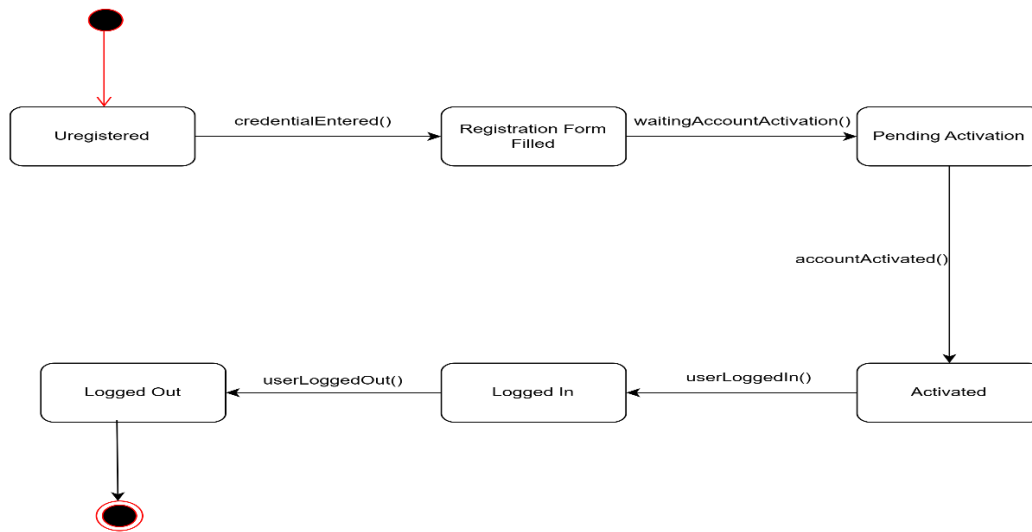
#### 4. Communication diagram to authenticate



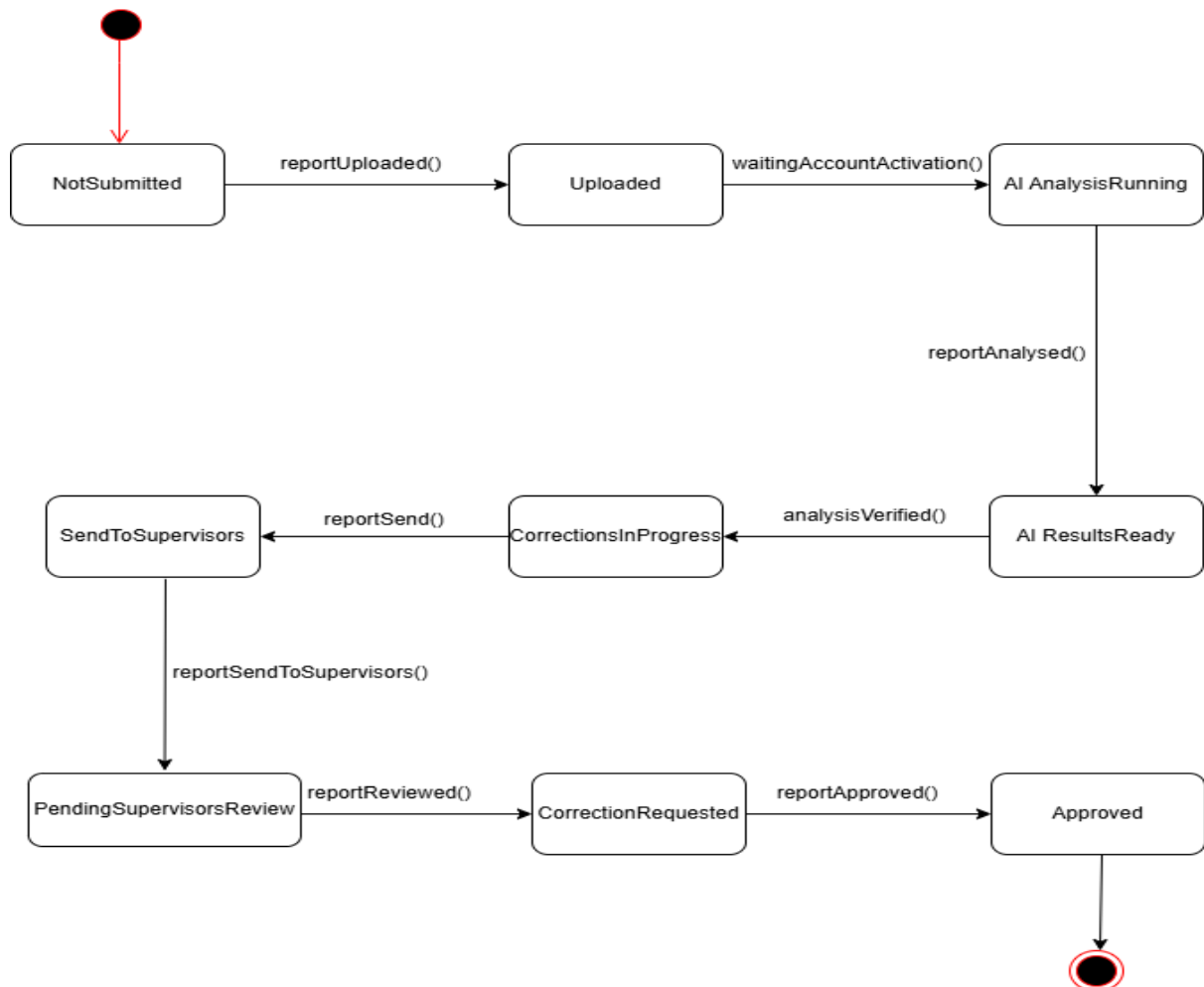
#### 4.1 communication diagram to upload report



## 5. State machine diagram for user account

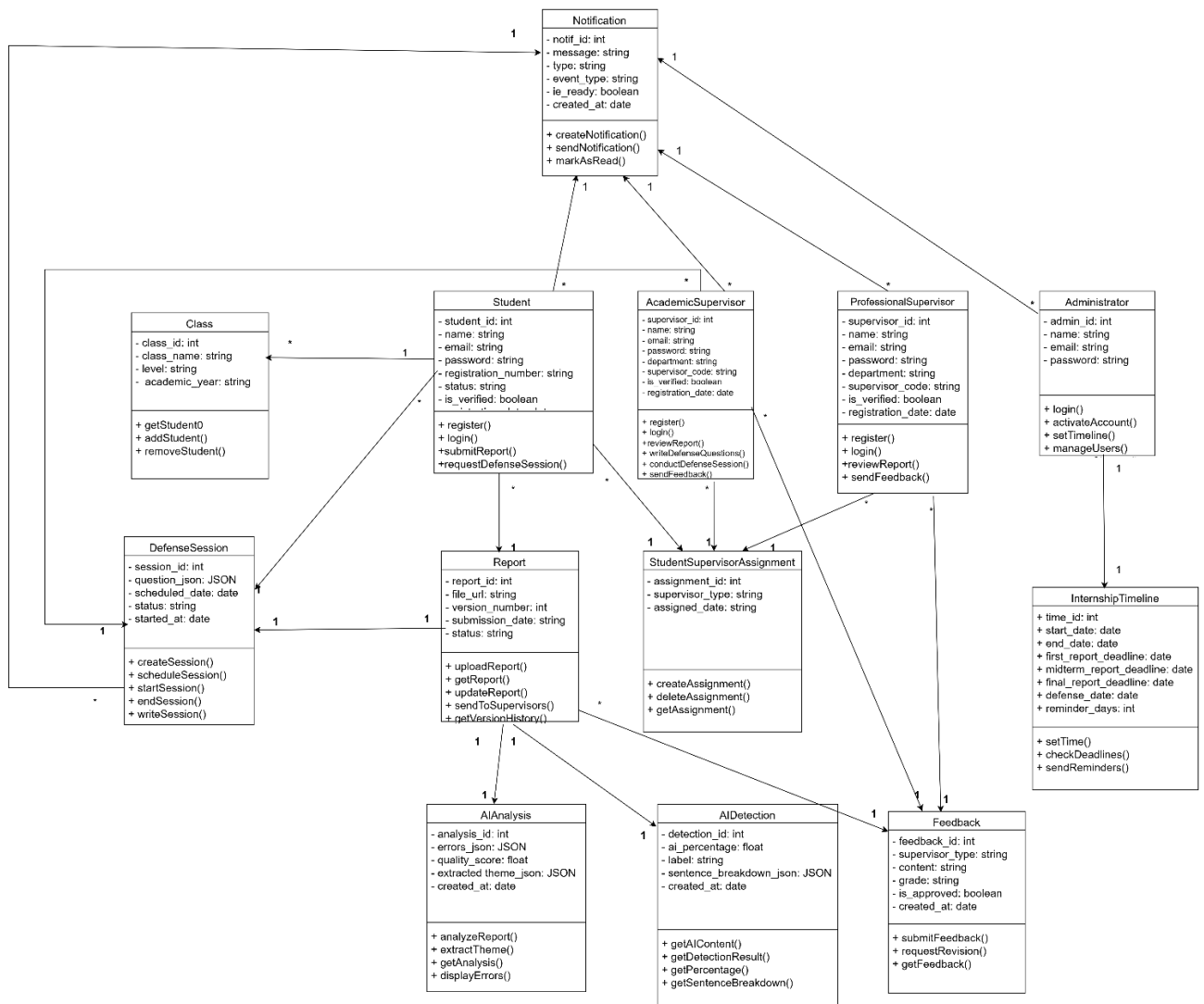


### 5.1 Report state machine diagram

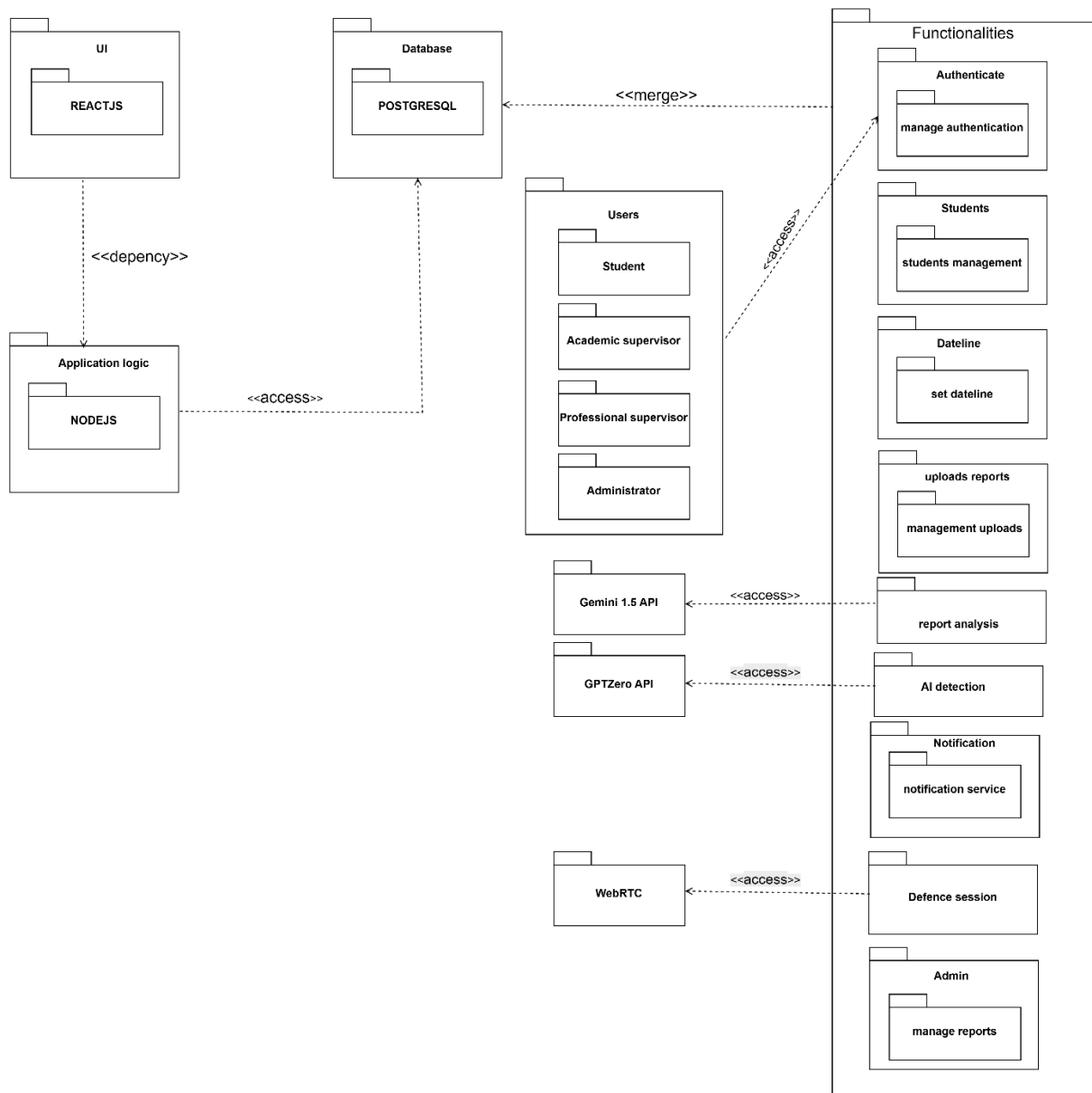




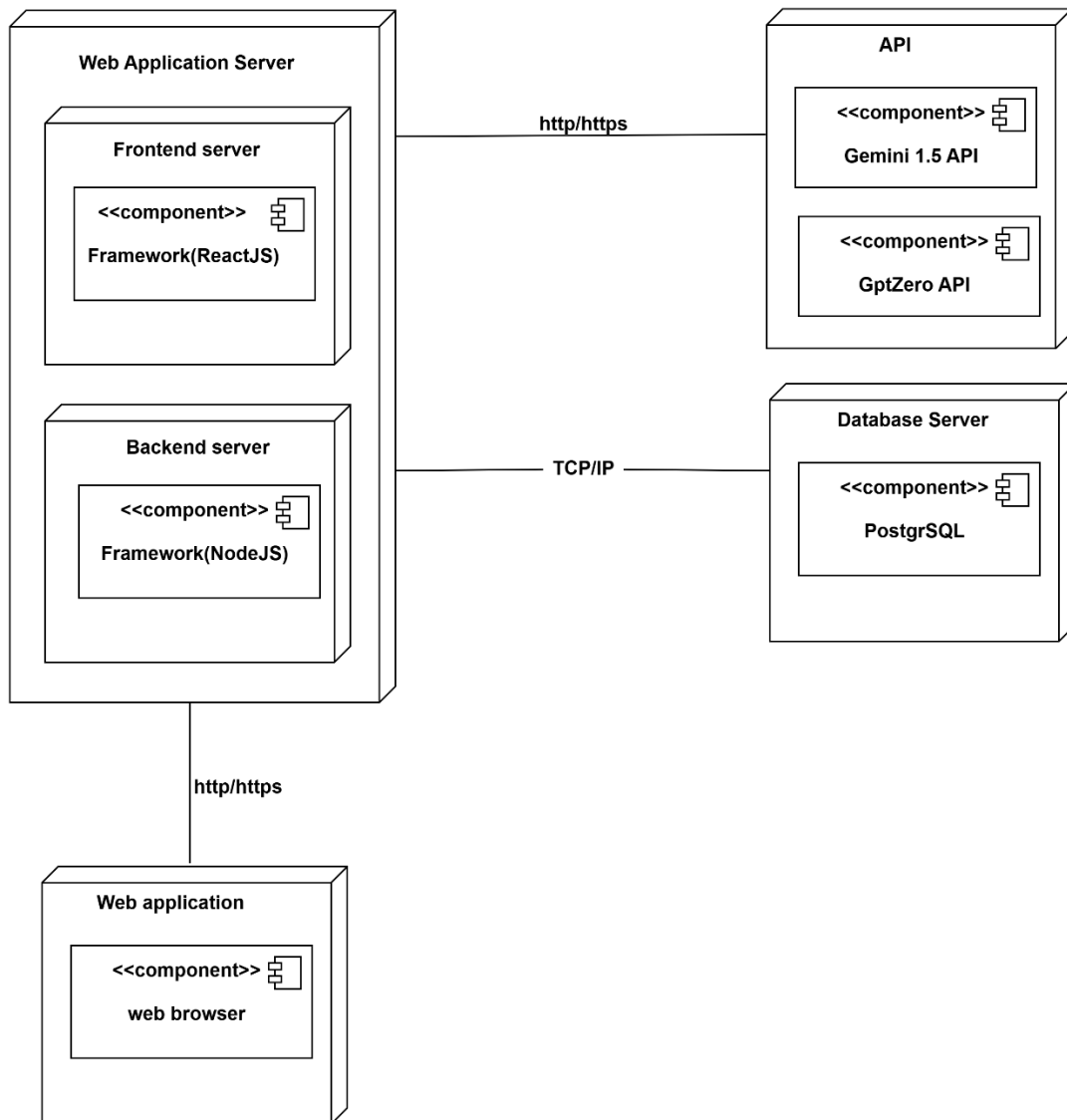
## 6. Class diagram



## 7. Package diagram



## 8. Deployment diagram



## 9. Component diagram